Bank Marketing Subscription Prediction System

Academic Machine Learning Evaluation Report | Generated: 2026-10-25 17:49:40

1. Executive Summary & Problem Formulation

Direct telemarketing campaigns in the banking sector incur substantial operational expenditures while typically yielding modest conversion rates. This project develops an end-to-end Machine Learning system to classify whether prospective bank clients will subscribe to a term deposit. To enforce methodological rigor and prevent feature leakage, call duration is intentionally omitted, ensuring realistic pre-contact predictive efficacy.

2. Dataset & Experimental Setup

Parameter	Configured Value	Parameter	Configured Value
Benchmark Dataset	bank-full.csv	Total Records	45,211
Training Split (80%)	36,168	Test Split (20%)	9,043
Target Variable	y ('no' / 'yes')	Positive Class (yes)	11.70% (Imbalanced)
Leakage Prevention	Excluded 'duration'	Validation Protocol	Stratified Test Partition

3. Empirical Four-Model Benchmark

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Train (s)
Logistic Regression	89.33%	66.32%	17.86%	28.15%	77.17%	0.36s
Decision Tree Classifier	89.40%	63.27%	22.31%	32.98%	71.48%	0.29s
K-Nearest Neighbours (KNN)	89.26%	65.16%	17.67%	27.81%	73.69%	0.10s
Random Forest Classifier	89.47%	69.49%	17.86%	28.42%	80.47%	1.36s

4. Technical Findings & Class Imbalance Analysis

Across the held-out evaluation subset of 9,043 instances, **Random Forest Classifier** attained the highest overall discriminative power with an ROC-AUC of 80.47%, while **Decision Tree Classifier** delivered top balanced harmonic score with an F1 of 32.98%. Due to the pronounced 88.3% / 11.7% class imbalance, accuracy alone is misleading: a naive dummy classifier predicting zero universally achieves 88.30% accuracy while failing completely on all subscribers. The balanced class-weighting framework deployed across our ensemble and linear models elevates recall for the subscription class, ensuring valuable prospect discovery for telemarketing campaign managers.

5. Verified Literature Review (2021-2024)

Study & Authors	Venue / DOI	Key Findings & Alignment
Alessa et al. (2023) ICCMA	IEEE Xplore 10.1109/ICCMA59762.2023.10374595	Evaluated LR, DT, KNN, and RF on Bank Marketing; ensemble learners demonstrated s
Singh et al. (2024) TIACOMP	IEEE Xplore 10.1109/TIACOMP64125.2024.00085	Demonstrated the crucial importance of addressing class imbalance; resampling improv
Dixit & Kokila (2024) ICONAT	IEEE Xplore Doc: 10700543	Benchmarked diverse classifiers and ensemble stacking, achieving ~90% accuracy with

Tanvir et al. (2024) DASA / arXiv	IEEE / arXiv 10.48550/arXiv.2410.21539	Probabilistic Bayesian regression provides calibrated certainty intervals for marketing re
Khan et al. (2021) ICCSA	Springer LNCS 10.1007/978-3-030-87010-9_8	Highlighted customer balance and contact history importance; emphasized explainable

6. Ethical Considerations & System Limitations

Notice: *Model predictions represent probabilistic estimates based on historical observational records. Outputs should augment human decision-making rather than act as autonomous credit or marketing rejection mechanisms. Feature correlations must not be interpreted as definitive causal proof.*